No Lead Beats Bedaquiline on Its Own Site: A Vina Benchmark of Five ATP Synthase Inhibitors on the M. tuberculosis Fo Structure 8J57

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Author: Zubair Ali, Pharm-D student, Shaheed Mohtarma Benazir Bhutto Medical University (SMBBMU), Larkana, Pakistan **Drafting assistant:** Lapis (AI) **Date:** 2026-11-07 · **Health Papers Agent — Paper #4** (daily 5-paper run)

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Abstract

Tuberculosis made about 10.8 million people ill in 2023 and killed about 1.25 million, and Pakistan alone accounted for about 686,000 cases and 47,000 deaths. Bedaquiline, the first diarylquinoline, blocks the Mycobacterium tuberculosis ATP synthase by binding its c-ring subunit AtpE, but it carries boxed warnings for QT prolongation and increased mortality, and resistance is already emerging through atpE target mutations and mmpR5 efflux derepression. We ran the LeadForge pipeline live on 7 November 2026 (UniProtKB → ChEMBL → PubChem → RDKit → PAINS/ADMET → Vina docking) against the M. tuberculosis ATP synthase Fo structure in complex with bedaquiline (PDB 8J57, 2.85 Å cryo-EM), with the grid centred on the co-crystallised bedaquiline copy at the a–c interface, and docked bedaquiline plus five known ATP synthase inhibitors with measured on-target IC50 values of 1–70 nM, all under an identical AutoDock Vina 1.2.7 protocol. No lead beat bedaquiline: bedaquiline scored best at −8.13 kcal/mol, the closest lead (CHEMBL3754231, IC50 10 nM) trailed by 0.55 kcal/mol, and the most potent measured inhibitor (CHEMBL3754673, IC50 1 nM) scored only −7.37 kcal/mol — every margin inside Vina's ±1–2 kcal/mol noise, and the docking order disagrees with the measured potency order. The falsifiable next step is a head-to-head ATP synthesis inhibition assay of the best-docking lead against bedaquiline on wild-type and Glu61Asp AtpE enzyme, with a pre-registered rule that a docking margin under 2 kcal/mol predicts no measurable advantage; a lead that keeps potency on the Glu61Asp mutant, or a measured IC50 advantage over bedaquiline, would refute the claim that the docking ranking is only a measurement of Vina's limits.

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1. Introduction

Tuberculosis (TB) remains one of the world's deadliest infectious diseases: according to the WHO Global Tuberculosis Report 2024, about 10.8 million people fell ill with TB in 2023 and about 1.25 million died. Pakistan is the fifth-highest-burden country, with an estimated 686,000 incident cases in 2023 (6.3% of the global burden), about 47,000 deaths, and about 15,000 people developing rifampicin-resistant TB. Drug-resistant TB needs longer, more toxic regimens, so new drugs with new mechanisms remain urgently needed.

Bedaquiline (BDQ, Sirturo, R207910/TMC207) is the first-in-class diarylquinoline, approved by the US FDA in December 2012 for multidrug-resistant TB — the first new TB drug class in four decades. It kills M. tuberculosis by blocking ATP synthase, the rotary enzyme that makes ATP from the proton gradient: diarylquinolines bind the oligomeric c-ring subunit AtpE in the membrane Fo domain and jam the rotor. The target was established genetically and biochemically in 2007, the isolated M. phlei c-ring was crystallised with bedaquiline at 1.7 Å in 2015, the M. smegmatis enzyme was solved by cryo-EM in 2021, and in 2024 the M. tuberculosis enzyme itself was solved with bedaquiline and with

its analogue TBAJ-587 bound — three binding sites in the Fo domain, with the tightest at the interface of subunit a and the c-ring.

Bedaquiline's success is shadowed by two problems this paper is built around. First, safety: the FDA label carries boxed warnings for QT prolongation (bedaquiline blocks the cardiac hERG potassium channel; ECG monitoring is required and the drug is stopped if QTcF exceeds 500 ms) and for increased mortality (11.4% versus 2.5% on placebo in one trial). Bedaquiline also inhibits the human mitochondrial ATP synthase in vitro, with an IC50 of about 0.66 μM — the same binding site, in the human enzyme, at a concentration within reach of clinical exposure. Second, resistance: the WHO mutation catalogue associates only three genes with bedaquiline resistance — atpE, mmpR5 (Rv0678) and pepQ. Efflux-based resistance through mmpR5 derepression of the MmpS5–MmpL5 pump is the common route and usually gives a modest 2- to 8-fold MIC rise; target-based atpE mutations are rarer but give higher MICs and are thought to emerge after the efflux step. In vitro passaging shows the sequence clearly: dynamic rv0678 mutations first (MIC about 1 μg/mL), then fixed atpE mutations such as Glu61Asp (MIC 2 to >2 μg/mL). In clinical isolates, atpE mutations remain rare — only 17 clinical isolates with atpE mutations had been described in nine studies before a 2025 Russian series found Glu61Asp or Ile66Met in 4 of 8 bedaquiline-resistant strains — but they sit exactly at the drug-binding interface, at residues that contact bedaquiline directly.

The second-generation diarylquinolines show the class can be improved on both axes: TBAJ-587 is about four times more potent than bedaquiline against H37Rv (MIC 0.016 versus 0.0625 μg/mL) and eight times more potent against an Rv0678 mutant, and its first-in-human trial reported no clinically significant QTc effects; TBAJ-876 is at least tenfold more potent in MIC assays against diverse strains, including Rv0678 mutants, with reduced QTc risk in preclinical and phase 1 studies. The question this paper asks is narrower and methodological: can a standardised, automated docking pipeline rank known ATP synthase inhibitors against bedaquiline on the true M. tuberculosis structure, and what does an apparent docking "win" mean when the reference drug's own binding site is the grid? The falsifiable hypothesis is that any lead beating bedaquiline by less than 2 kcal/mol on Vina docking has no measurable potency advantage, and that the docking ranking will disagree with measured ATP synthase IC50 values on this target.

2. Methods

2.1 Literature basis

Sources were searched on 7 November 2026: the daily literature digest at `~/workspace/research-team/digests/2026-11-07.md` (no bedaquiline item that day; the digest's AMR-generalizability review is cited for the validation framing), PubMed and web search, ChEMBL (EBI), RCSB PDB and UniProtKB. Key sources: the WHO Global Tuberculosis Report 2024 for burden figures; the 2007 Nature Chemical Biology paper establishing AtpE as the diarylquinoline target; the 2024 Nature paper solving the M. tuberculosis ATP synthase with bedaquiline and TBAJ-587; the FDA Sirturo label (DailyMed, updated 17 October 2025) for the boxed warnings; the 2019 Antimicrobial Agents and Chemotherapy stepwise-resistance study; the 2025 Emerging Infectious Diseases report on atpE mutations in Russian clinical isolates; and the TBAJ-587/TBAJ-876 phase 1 and mouse-efficacy reports. All ChEMBL potency values quoted were re-checked live against the ChEMBL API on 7 November 2026 for the assay "Inhibition of Mycobacterium tuberculosis ATP synthase".

2.2 Molecular docking

The LeadForge pipeline was run live on 7 November 2026 (UniProtKB → ChEMBL → PubChem → RDKit → PAINS/ADMET → Vina). Two pipeline faults were caught and corrected before reporting, and both are documented here because they are part of the result.

Fault 1 — wrong UniProt entry. The pipeline resolved the query "Mycobacterium tuberculosis ATP synthase subunit c AtpE" to UniProt P9WPV1, which is the ATP synthase **epsilon chain** (gene atpC, Rv1311, 121 aa) — not AtpE. The task brief's own suggested accession, P9WPV3, is also wrong: it is the ATP synthase subunit b-delta (gene atpFH, Rv1307). The correct AtpE entry was verified live against UniProtKB on 7 November 2026: **P9WPS1** (ATPL_MYCTU, gene atpE, Rv1305, 81 aa). The ChEMBL side was correct: target CHEMBL2364166, "ATP synthase", a PROTEIN COMPLEX of M. tuberculosis with 8 components, from which 17 bioactivities were retrieved. The docking structure was therefore chosen by hand, not by the pipeline's UniProt lookup: PDB **8J57**, "Cryo-EM structure of Mycobacterium tuberculosis ATP synthase Fo in complex with bedaquiline (BDQ)", 2.85 Å, from Zhang et al., Nature 2024 — whose chains 1–9 are subunit c (AtpE, P9WPS1) and chain a is subunit a.

Fault 2 — no PDB found. The pipeline's structure picker only accepts X-ray structures better than 3.0 Å, so it returned `no_pdb_found` and skipped docking entirely: the only M. tuberculosis ATP synthase structures with bedaquiline bound are cryo-EM (8J0S full complex, 8J57 Fo). Docking was therefore run manually with the same LeadForge `dock()` function and protocol.

Docking was performed with AutoDock Vina 1.2.7, using the LeadForge protocol: the same structure, grid centre, box (22.5 Å cube), exhaustiveness (8) and pose count (3) for every ligand; best of three poses reported. The grid was centred on the centroid of the co-crystallised bedaquiline copy on **chain 1** of 8J57: **[162.46, 186.93, 275.26]**. The centre was verified against the PDB file: 8J57 holds seven bedaquiline copies (ligand code BQ1, residue 600) on c-ring chains 1, 4, 5, 6, 7, 8 and 9. The chain-1 copy sits at the a–c interface — its centroid is 5.13 Å from the nearest subunit-a atom and its closest atom contact to subunit a is 3.04 Å — while the chain-4 copy is the second a-contacting site (6.09 Å) and the remaining five copies are c-only sites 13.5–37.6 Å from subunit a. The chain-1 copy is therefore one of the two tight a–c interface sites described by Zhang et al., not a cofactor, glycan or lipid: the grid-centre fault that has corrupted earlier runs (centring on a cofactor or glycan) did not occur here. Receptor preparation used RDKit (Gasteiger charges, simplified AutoDock atom types, HETATM ligands stripped); ligand preparation used meeko. Vina's scoring noise is approximately ±1–2 kcal/mol; differences smaller than this cannot be treated as significant on scores alone.

Six ligands were docked: bedaquiline itself (QUERY, PubChem CID 5388906) and the top five pipeline-ranked leads, all known ATP synthase inhibitors with ChEMBL potencies measured in the on-target enzyme assay. All five leads are diarylquinoline analogues except CHEMBL4101131, a squaramide (cyclobutenedione) chemotype with low structural similarity to bedaquiline (Tanimoto 0.073).

2.3 PAINS screen

Every test compound was screened with the RDKit PAINS filter: bedaquiline and all five leads were PAINS-clean (no alerts). A ToxGuard Tox21 kNN read-across panel (12 assays) was run for each lead. The highest read-across probabilities were NR-AhR 0.179 for CHEMBL3754673, SR-MMP 0.134 for CHEMBL3754231, SR-ARE 0.245 for CHEMBL4101131, and NR-ER-LBD 0.086 for bedaquiline itself. These are similarity-based predictions, not measurements.

2.4 Honesty limits

What this paper cannot show: docking is not activity; Vina noise is ±1–2 kcal/mol; no experimental data are presented; no molecular-dynamics refinement, binding-free-energy calculation, or re-docking validation of the co-crystallised bedaquiline pose (RMSD) was performed; a single cryo-EM structure (8J57, one Fo conformation, 2.85 Å) was used; only one of the three bedaquiline binding sites (the a–c interface site) was docked, and the c-only and lagging sites were not scored; and the leads are already-known ATP synthase inhibitors, so no novelty of chemistry is claimed. Bedaquiline's QT-prolongation liability (hERG block) was not modelled for any lead, so a docking win says nothing

about cardiac safety.

3. Results

3.1 Docking: the comparison table

Compound	ATP synthase IC50, ChEMBL	Best affinity (kcal/mol)	Poses (kcal/mol)	MW	logP	Lipinski violations	PAINS	Top tox read-across
Bedaquiline (reference, query)	—	**−8.13**	−8.13, −7.91, −7.45	555.52	7.13	2	clean	NR-ER-LBD 0.086
CHEMBL3754231	IC50 10 nM	−7.58	−7.58, −7.35, −7.31	586.53	6.53	2	clean	SR-MMP 0.134
CHEMBL3754673	IC50 1 nM	−7.37	−7.37, −7.11, −7.10	557.69	6.92	2	clean	NR-AhR 0.179
CHEMBL3751877	IC50 50 nM	−7.03	−7.03, −6.84, −6.84	586.53	6.53	2	clean	NR-ER 0.090
CHEMBL3752792	IC50 70 nM	−6.85	−6.85, −6.53, −6.41	507.63	5.77	2	clean	NR-ER 0.092
CHEMBL4101131 (squaramide)	IC50 30 nM	−6.14	−6.14, −5.86, −5.68	349.39	1.79	0	clean	SR-ARE 0.245

All docking on PDB 8J57 (M. tuberculosis ATP synthase Fo + bedaquiline, 2.85 Å cryo-EM), grid centre [162.46, 186.93, 275.26] (chain-1 BQ1 centroid, a–c interface site), box 22.5 Å, exhaustiveness 8, Vina 1.2.7, 7 November 2026. All five leads are known ATP synthase inhibitors; IC50 values are ChEMBL on-target enzyme-assay potencies, re-checked live on 7 November 2026.

The headline, stated honestly: **no lead beats bedaquiline**. Bedaquiline scores best at −8.13 kcal/mol with a tight pose cluster (−8.13, −7.91, −7.45; 0.68 kcal/mol spread), and every lead trails it — the closest by 0.55 kcal/mol, the furthest by 1.99 kcal/mol. Every margin is inside Vina's ±1–2 kcal/mol noise, so not even the ranking of the leads can be treated as significant, let alone a "win" over the reference. The docking order also disagrees with measured potency: by IC50 the order is 1, 10, 30, 50, 70 nM, but by docking it is bedaquiline, then the 10 nM, 1 nM, 50 nM and 70 nM compounds, with the 30 nM squaramide last. The most potent measured inhibitor in the set (CHEMBL3754673, IC50 1 nM) scores third at −7.37 kcal/mol, 0.76 kcal/mol behind bedaquiline. The docking scores span only 1.99 kcal/mol while the measured potencies span 70-fold — Vina compresses the potency range on this site and ranks within it unreliably. The squaramide CHEMBL4101131 is the instructive failure: a different chemotype (Tanimoto 0.073 to bedaquiline), the only lead with zero Lipinski violations and the lowest logP (1.79), and the worst docking score (−6.14) despite a respectable 30 nM IC50 — either the diarylquinoline-shaped site genuinely disfavours it, or Vina cannot score a non-diarylquinoline in a diarylquinoline pocket. Either way, its score is not evidence against its potency. Pose consistency does not rescue the ranking: every compound's three poses sit within 0.68 kcal/mol of its best score, so the scores are stable — stably wrong in potency order.

3.2 Comparison with published docking

No published Vina score for bedaquiline or any of these leads on 8J57 was found to benchmark against — which is itself a gap this paper fills: Vina 1.2.7 reference values for bedaquiline and five known ATP synthase inhibitors on the true M. tuberculosis Fo structure are now on record for others

to reproduce, together with the corrected-centre audit trail. The ordering is not comparable to published potency: the five leads span measured ATP synthase IC50 values of 1–70 nM, a 70-fold spread, while their docking scores span 1.99 kcal/mol in an order that disagrees with potency — the 1 nM compound third, the 30 nM squaramide last. Vina ranks poses, not IC50s, and on this site it ranks them unreliably.

3.3 The standardisation / quality problem

Two things that could silently corrupt this result did not, and two did. The grid centre was verified against the PDB file to 0.01 Å and sits on a bedaquiline copy at the a–c interface, 3.04 Å from subunit a — the cofactor/glycan-centring fault is absent here, because the structure holds no cofactor at all. The pipeline's UniProt resolution fault did occur: left uncorrected, the paper would have been written against the epsilon chain (P9WPV1) while claiming to target AtpE, and the task brief's own suggested accession (P9WPV3, subunit b-delta) would have been equally wrong. The pipeline's X-ray-only structure picker also failed silently, skipping docking with `no_pdb_found` on a target whose only liganded structures are cryo-EM. The deeper problem is structural, not procedural: bedaquiline binds three sites in the Fo domain, and docking one site — even the tightest — cannot represent the drug's full binding energetics, which include the conformational change the drug induces in the enzyme.

4. Discussion

4.1 The reframe

The result reframes the paper from a discovery claim into a negative benchmark with a useful control. There is no lead here: bedaquiline beats all five known inhibitors on its own site, the closest by 0.55 kcal/mol — inside noise — and the most potent measured inhibitor (1 nM) is only third on docking. The durable findings are the benchmark values for bedaquiline on 8J57 (–8.13 kcal/mol, with full pose lists), the demonstration that Vina compresses a 70-fold potency range into 1.99 kcal/mol and orders it wrong, and the audit trail of two pipeline faults caught before reporting — the wrong-UniProt-entry fault and the X-ray-only structure-picker fault — both of which would have produced a confidently wrong paper. The squaramide result is the most actionable negative: a non-diarylquinoline with a 30 nM on-target IC50, zero Lipinski violations and the best drug-likeness in the set scores worst on docking, which is exactly the profile a diarylquinoline-shaped docking screen would discard — and exactly the profile that might escape both bedaquiline's hERG liability and the atpE resistance mutations that sit in the diarylquinoline pocket.

4.2 Why the "win" must not be over-read

Three reasons. First, any margin under 2 kcal/mol is inside Vina's ±1–2 kcal/mol noise. Second, the ranking must be checked against measured ATP synthase IC50 values — the five leads span 1–70 nM, and a docking order that disagrees with that order is a measurement of the scoring function, not of the molecules. Third, only one of bedaquiline's three Fo binding sites was scored, and the cryo-EM structure is a single Fo conformation; the drug's potency comes partly from the large conformational change it induces, which a rigid-receptor docking cannot model. A docking "win" against bedaquiline on one site is not a discovery; it is a benchmark value awaiting an assay.

4.3 The PAINS problem is the experiment's design constraint, not a reason to stop

All leads are PAINS-clean, which removes one artefact class from the validation design — but PAINS-clean is a low bar, and the Tox21 read-across flags (up to SR-ARE 0.245) are predictions from neighbours, not data. The honest reading: nothing in the panel kills any lead, and nothing in it advances one either. The validation assay must still include redox controls and a counter-screen, because a clean PAINS filter does not rule out assay interference. The larger safety question — hERG block and QT prolongation, the class liability that carries a boxed warning for bedaquiline — is not addressed by PAINS at all and needs a dedicated hERG assay for any lead that advances.

4.4 Testable hypotheses

• The best-docking lead inhibits recombinant M. tuberculosis ATP synthase with an IC50 within 2-fold of its ChEMBL value under identical assay conditions; a measured IC50 worse than bedaquiline's would confirm the noise interpretation, while a better one would refute it.

• A lead that docks well at the a–c interface site keeps its potency on Glu61Asp AtpE enzyme, the atpE mutation found in 4 of 8 bedaquiline-resistant Russian clinical isolates; loss of potency on the mutant would show the docking site is the resistance site, not a way around it.

• A Vina re-dock of the co-crystallised bedaquiline (chain-1 BQ1) into 8J57 reproduces its crystallographic pose within 2 Å RMSD (protocol self-validation; not yet run).

• Docking the same six ligands at the second a-contacting site (chain-4 BQ1 copy) and at a c-only site (chain-6 BQ1 copy) gives the same ranking as the chain-1 site; a different ranking would show the result is site-dependent, not molecule-dependent.

4.5 Proposed validation (university-lab feasible, 8–12 weeks)

Stage 1 (weeks 1–2): ATP synthesis inhibition assay (the same luminometer-based assay reported for CHEMBL4101131's 30 nM IC50) for the best-docking lead, the second-best lead and bedaquiline, same plate, n = 3, with a pre-registered 2-fold decision rule and the rule that a docking margin under 2 kcal/mol predicts no measurable advantage. Stage 2 (weeks 3–6): the same assay on Glu61Asp and Ile66Met AtpE mutant enzyme for any compound that clears Stage 1, to test whether the lead survives the two atpE mutations most often seen in resistant clinical isolates. Stage 3 (weeks 7–12): whole-cell M. tuberculosis H37Rv MIC assay for any compound that clears Stage 2, with bedaquiline as control, plus a hERG channel assay, since QT prolongation is the class liability that a docking score cannot see. Total compound cost is the binding constraint; all leads are commercially available research chemicals, not approved drugs, and none may be presented as a treatment.

5. Limitations

• Docking only; no experimental data. 2. Any margin under 2 kcal/mol is inside Vina's ±1–2 kcal/mol noise and cannot be interpreted as a real advantage. 3. Only one of bedaquiline's three Fo binding sites was docked; the c-only and lagging sites were not scored. 4. No molecular-dynamics refinement, binding-free-energy calculation, or re-docking RMSD validation of the co-crystallised bedaquiline pose was performed. 5. A single cryo-EM structure (8J57, one Fo conformation, 2.85 Å) was used; the drug-induced conformational change is not modelled and the receptor was held rigid. 6. All leads are already-known ATP synthase inhibitors; no new chemical matter is proposed. 7. The pipeline's UniProt resolution picked the wrong subunit (epsilon chain instead of AtpE) and its structure picker skipped cryo-EM structures entirely — other silent defaults in the same pipeline are not yet audited. 8. The hERG/QT liability was not modelled for any lead. 9. The resistance analysis is literature-based: no mutant enzyme was docked and no MIC was measured.

6. Conclusion

No ATP synthase inhibitor in this set beats bedaquiline on its own binding site: bedaquiline scores −8.13 kcal/mol on the M. tuberculosis Fo structure 8J57, every lead trails it by 0.55–1.99 kcal/mol — inside Vina's noise — and the docking order disagrees with measured potency, with the 1 nM inhibitor third and the 30 nM squaramide last. The honest reading is that on this membrane site, non-covalent docking cannot rank known inhibitors against bedaquiline — and the next experiment is a head-to-head ATP synthesis assay on wild-type and Glu61Asp AtpE enzyme with a pre-registered 2 kcal/mol rule, plus a hERG assay for any lead that advances, not more Vina docking.

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Data availability

All docking scores reported here are in `hidden_files/dock-HP-020c-2026-11-07.json` (manual dock on 8J57) and `hidden_files/dock-HP-020-2026-11-07.json` (pipeline run: leads, ADMET/PAINS, no docking — structure picker skipped cryo-EM). ChEMBL IDs: bedaquiline (query, PubChem CID 5388906), CHEMBL3754673, CHEMBL3751877, CHEMBL3754231, CHEMBL3752792, CHEMBL4101131. Target: UniProt P9WPS1 (AtpE), ChEMBL CHEMBL2364166. PDB ID 8J57. No additional data were generated.